Gated Cluster-Pooled Sign Constraints for Multi-Output Sparse Regression

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Abstract

Marginal sign restrictions stabilise sparse multi-output fits, but a sign estimated one response at a time is noisy when observations are few. We pool marginal slopes across clusters of responses, scale each pooled slope by a date-level sandwich variance that retains contemporaneous response dependence, and use the resulting statistic as a gate that restricts a coefficient's direction, excludes the coefficient, or leaves it free before a sparse-group Lasso fit. For fixed dimensions we give sufficient conditions under which the gate recovers the direction of every nonzero coefficient, a checkable condition that implies the required marginal-sign agreement, and the precision ratio $\sqrt{m/\{1+(m-1)\rho_e\}}$ of a slope pooled over m responses with marginal-residual correlation ρ_e, which a Monte Carlo check confirms to within one percent. In simulations with learned clusters, pooling raises correct active signs at null-cell retention **0.10** by **0.030** when cluster members share supports and directions, and lowers them by **0.066** and **0.134** under mixed directions and partial supports. With exclusion it raises test R^2 in the coherent design from **0.614** to **0.626**, level with cooperative Lasso and UniLasso. In repeated nested validation on yeast expression quantitative trait locus data, excluding failed gates at fixed settings lowers median-gene R^2 from **0.073** to **0.035** while releasing them leaves it at **0.073**. With cut and threshold tuned jointly with the penalty, exclusion moves to near-singleton partitions and the loss falls to **0.007** without a gain over unconstrained fitting. The gate should restrict directions. Exclusion pays only when the grouping is coherent.

Keywords: Marginal screening, Response dependence, Variable selection, Regularization, Nested validation, Gene expression

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1 Introduction

In multi-output regression, a coefficient's sign describes the direction of a conditional association. Examples include whether a factor loading increases portfolio exposure (Sepp et al. 2026) and whether a genetic marker is associated with higher or lower expression after adjusting for other markers. A marginal regression can estimate a different direction when predictors are correlated. Pooling marginal evidence across responses can improve precision, but it does not resolve that distinction. We study when a pooled marginal sign is a useful restriction for a subsequent sparse regression, and when it excludes a nonzero coefficient or constrains it in the wrong direction.

The univariate-guided least absolute shrinkage and selection operator (LASSO) of Chatterjee et al. (2025) is the incumbent construction. It restricts each coefficient to the sign of its own marginal regression and fits a nonnegative second stage on univariate features. The pooled gate replaces the per-response sign by a cluster-level sign and adds a screening step. The cost of switching is a response partition and a threshold, both of which must be tuned. Sign constraints alone regularise a regression: Meinshausen (2013) and Slawski and Hein (2013) show that sign- and nonnegativity-constrained least squares recover sparse signals without a penalty under conditions on the design. Those results take the signs as given, whereas the gate supplies them. The cooperative LASSO of Chiquet et al. (2012) encourages sign coherence through a penalty that learns the group direction during fitting and can set groups to zero. We instead fix a permitted direction, or exclude a coefficient, before the penalized fit. Group and multivariate sparse group penalties (Yuan and Lin 2006; Obozinski et al. 2010; Li et al. 2015) encode coefficient groups, multivariate regression with covariance estimation (Rothman et al. 2010) and structured penalties for eQTL mapping (Peng et al. 2010; Kim and Xing 2012) share information across responses through the loss or the penalty, and adaptive penalties (Zou 2006; Wang and Leng 2008) use pilot magnitudes to reweight shrinkage. Marginal screening (Fan and Lv 2008) supplies the idea of filtering weak marginal evidence before fitting. We compare the complete fitted estimators of these families as well as their preliminary screens.

We make three contributions. We develop a dependence-aware pooled sign gate whose statistic scales the cluster-pooled slope by a date-level score variance, and whose failure can either exclude a coefficient or leave it free. We establish, for fixed dimensions, sufficient conditions under which the gate retains the correct direction on every nonzero coefficient (Proposition 1), give a checkable condition on predictor covariances and coefficients that implies the required marginal-sign agreement (Lemma 1), and derive the precision ratio $\sqrt{m/\{1+(m-1)\rho_e\}}$ of pooling m responses whose marginal residuals have correlation ρ_e, which a Monte Carlo check confirms to within one percent (Fig. 1). We show in which response structures the gate helps and in which it harms. At a null-cell retention of 0.10, pooling changes the proportion of correct active signs by $+0.030$ in a coherent-group design and by -0.066 and -0.134 under mixed directions and partial supports (Fig. 2). In repeated nested validation on yeast eQTL data, excluding failed gates at fixed settings lowers median-gene R^2 from 0.073 to 0.035, and tuning the cut and threshold with the penalty removes the loss without producing a gain (Fig. 3). The theory concerns gate directions on nonzero coefficients

and not the support of the fitted coefficient matrix. The experiments establish no general predictive advantage over cooperative Lasso, multivariate sparse group Lasso or UniLasso.

We evaluate complete fitted estimators in two simulation families. The first compares nine methods across six low- and moderate-signal designs. The second uses 40 paired replications of five recognizable-group designs, jointly tunes cut, mixing, threshold and penalty where applicable, and measures actual partition recovery. The yeast eQTL analysis repeats preprocessing within nested validation and measures prediction, partition stability, coefficient selection and conditional direction agreement. Extended results and proof details are supplied in Online Resource 1, and Online Resource 2 contains the replication source and numerical evidence. The method is implemented in the open-source Python package `factorlasso`.[1] The broader software interface and portfolio applications are described in an unpublished companion manuscript by the same authors.

Section 2 develops the method. Section 3 states the on-support gate result, the sufficient condition for marginal-sign agreement and the pooling precision calculation. Section 4 reports the simulation study. Section 5 applies the method to yeast eQTL data. Section 6 describes computation and reproducibility, and Section 7 discusses the implications.

2 Methodology

The procedure takes a training matrix of predictors and responses, a response partition or clustering rule, a gate threshold τ, a variance convention, and penalty parameters (λ, α). Its outputs are the pooled slopes, their screening statistics, the constraint matrix, and the fitted coefficient matrix. These are distinct evaluation targets.

1. Select complete observations and estimate any centering, scaling and predictor-screening transformations on the training data.
2. Supply response clusters or construct them from the training responses using the stated distance, linkage and cut rule.
3. Compute each cluster–predictor marginal slope and its date-score variance. The archived independent-cell variance is a separate comparator.
4. Retain the slope's direction when its statistic passes τ. Otherwise exclude that predictor for all cluster members by fixing the coefficients to zero, or leave the cells unrestricted under the release variant. Record any explicitly unrestricted cells separately.
5. Form the chosen hierarchical-clustering group LASSO (HCGL) or factor-clustering group LASSO (FCGL) penalty and solve the constrained optimization problem. Select hyperparameters using training-only validation, repeating Steps 1–4 within each inner training fold, then refit on the full outer training sample.

Held-out responses are used only for evaluation. The principal estimator uses complete, equally weighted observations. An optional masked and weighted gate is described

[1] https://github.com/ArturSepp/factorlasso

separately in Online Resource 1, Section S15. The remainder of this section defines each step.

2.1 Multi-response setup

Let $\boldsymbol{Y} \in \mathbb{R}^{T\times N}$ collect N responses observed over T observations, and let $\boldsymbol{X} \in \mathbb{R}^{T\times M}$ collect M predictors observed over the same observations. The multi-response regression model is

$$\boldsymbol{Y} \;=\; \boldsymbol{X}\,\boldsymbol{\beta}^{\top} + \boldsymbol{\varepsilon}, \tag{1}$$

where $\boldsymbol{\beta} \in \mathbb{R}^{N\times M}$ is the coefficient matrix and $\boldsymbol{\varepsilon} \in \mathbb{R}^{T\times N}$ is the residual. The estimator centers the columns of $\boldsymbol{Y}$ and $\boldsymbol{X}$ before fitting and recovers the intercept afterward. We write $\boldsymbol{x}_j \in \mathbb{R}^T$ for the j-th predictor column, $\boldsymbol{y}_k \in \mathbb{R}^T$ for the k-th response column, and β_{kj} for the (k, j) entry. We partition the N responses into clusters $\{\mathcal{C}_1, \ldots, \mathcal{C}_G\}$, where each $\mathcal{C}_g \subseteq \{1, \ldots, N\}$ is a set of response indices and $|\mathcal{C}_g|$ its size. We write $\mathcal{C}(k)$ for the cluster containing response k.

2.2 Cluster-pooled marginal slopes

For a response cluster $\mathcal{C}$ and predictor j, the pooled no-intercept least-squares slope is

$$\hat{\beta}_{j,\mathcal{C}} = \frac{\boldsymbol{x}_j^{\top} \sum_{k\in\mathcal{C}} \boldsymbol{y}_k}{|\mathcal{C}|\, \boldsymbol{x}_j^{\top} \boldsymbol{x}_j}. \tag{2}$$

It is the slope of the cluster-mean response on $\boldsymbol{x}_j$, extending the single-response marginal construction of Chatterjee et al. (2025). Broadcasting it gives every cluster member the same permitted direction. Broadcasting does not establish that their partial coefficients share that direction. Positive response correlation is insufficient, and a cluster can mix active and inactive responses.

2.3 The pooled sign gate

The pooled sign gate uses the date-score statistic $t_{j,\mathcal{C}} = \hat{\beta}_{j,\mathcal{C}}/\sqrt{\widehat{V}_{j,\mathcal{C}}}$, with the dependence-aware score scale defined in Section 2.4. The resulting sign matrix is

$$s_{kj} \;=\; \begin{cases} \operatorname{sign}\big(\hat{\beta}_{j,\mathcal{C}(k)}\big), & |t_{j,\mathcal{C}(k)}| \;\geq\; \tau, \\ 0, & |t_{j,\mathcal{C}(k)}| \;<\; \tau, \end{cases} \tag{3}$$

where $\tau \geq 0$ is a tuning parameter. Under exclusion, a failed gate sets $s_{kj} = 0$ and therefore forces $\beta_{kj} = 0$ for every $k \in \mathcal{C}$ on predictor j. Under release, a failed gate leaves those cells unrestricted. Exclusion can remove a genuine signal. A retained marginal sign can disagree with a partial coefficient and make that coefficient's true value infeasible, and a hard half-space still permits a nonzero estimate in that incorrect direction.

A fixed threshold supplies neither multiplicity control nor a guarantee of correct partial signs. Predictive tuning selects the threshold inside the training loop. The

eQTL analysis reports a fixed-threshold protocol and a jointly tuned one. Null-cell retention is distinct from the fraction of reported signs that are false.

2.4 Response dependence in the gate

The principal estimator uses complete observations with equal temporal weights. Write $\widehat{b}_j = \hat{\beta}_{j,\mathcal{C}}$ for the pooled slope defined above. Centering and scaling are learned on the training sample before clustering or pooling. For a cluster $\mathcal{C}$ of size m, define

$$D_j = m \sum_{t=1}^{T} x_{tj}^2, \qquad u_{tj} = x_{tj} \sum_{k \in \mathcal{C}} (y_{tk} - \widehat{b}_j x_{tj}), \qquad \widehat{V}_j = \frac{T}{T-1} \frac{\sum_t u_{tj}^2}{D_j^2}. \tag{4}$$

This score-sandwich construction retains contemporaneous marginal-residual covariance (White 1980; Liang and Zeger 1986; Cameron and Miller 2015). For a prespecified cluster and fixed analysis scales, it estimates the pooled slope's asymptotic variance under independent and identically distributed (iid) dates. When column standard deviations are learned from the same sample, it retains a positive score-scale limit under those conditions, which is sufficient for the on-support gate result, but we do not assert that it estimates the full sampling variance of the normalized slope. We use $t_j = \widehat{b}_j / \sqrt{\widehat{V}_j}$ in Equation (3). A predictor with $D_j = 0$ provides no slope evidence and is excluded. The finite-sample multiplier $T/(T-1)$ is a screening convention and not a claim of an exact Student distribution. Serial dependence would require an additional heteroskedasticity-and-autocorrelation-consistent (HAC) or block treatment.

The fitted protocols require at least two training observations and nonconstant training columns. If $\widehat{V}_j = 0$, a nonzero slope is assigned $t_j = \operatorname{sign}(\widehat{b}_j)\,\infty$ and a zero slope gives $t_j = 0$. For tuning, $\tau = +\infty$ is defined separately as complete gate failure, including these noiseless cases. The pooled slopes themselves are unchanged.

For a fixed cluster, repeating every member c times multiplies D_j and u_{tj} by c and leaves the slope and variance unchanged. Duplicating only one member of a heterogeneous cluster can change the estimand, and reclustering after duplication is outside this invariance statement. The balanced variance calculation in Section 3 uses marginal-regression residual correlation, not raw response correlation. The archived independent-cell comparator divides the pooled residual sum of squares by $(Tm-1)D_j$, which treats the m member residuals within a date as independent. Section 3 quantifies how far that comparator overstates precision.

2.5 Cluster discovery

The partition can be supplied externally or learned from training responses. The fitted experiments use Ward linkage on $D_{kl} = \sqrt{2(1 - C_{kl})}$, where $\boldsymbol{C} = \operatorname{corr}(\boldsymbol{Y})$. These are Euclidean distances between centered response vectors normalized to unit length when correlations use a common observation grid (Ward 1963; Murtagh and Legendre 2014). Arbitrary pairwise-missing correlations need not have this geometry. A cut fraction κ is applied to the final Ward merge height. The endpoints are implemented explicitly as all singletons and full pooling. Intermediate cuts, or an externally specified partition, need not yield coherent partial signs.

The first benchmark fixes $\kappa = 0.5$, the learned-group study tunes it inside training, and the eQTL baseline fixes $\kappa = 0.7$. The companion software manuscript's historical examples use a different correlation-distance heuristic, and their partitions are not interchangeable with those here.

2.6 Sign-constrained estimation

The derived signs constrain a multi-response sparse-group LASSO estimator (Simon et al. 2013). Let $\mathcal{U}$ be a specified set of unrestricted cells. On its complement the sign matrix has entries in $\{-1, 0, +1\}$, and Equation (3) defines the constraint set

$$\mathcal{K}(\boldsymbol{S}) = \left\{ \boldsymbol{\beta} : \begin{array}{rl} \beta_{kj} = 0 & \text{if } (k,j) \notin \mathcal{U},\ s_{kj} = 0, \\ s_{kj}\beta_{kj} \geq 0 & \text{if } (k,j) \notin \mathcal{U},\ s_{kj} \in \{-1,+1\} \end{array} \right\}. \tag{5}$$

A gated-out cell is pinned to zero, and a signed cell is held in the matching half-space. The pooled exclusion procedure has $\mathcal{U} = \varnothing$. The release variant moves the failed-gate cells into $\mathcal{U}$. An entirely unrestricted ablation takes $\mathcal{U}$ to be all cells, removing both zero-pins and sign half-spaces. Merely setting $\tau = 0$ keeps sign restrictions and is not an unrestricted fit. The software represents $\mathcal{U}$ by `NaN` entries. With all slopes and penalty weights held fixed after preprocessing, the HCGL estimator solves

$$\begin{aligned} \hat{\boldsymbol{\beta}} \;=\; \arg\min_{\boldsymbol{\beta} \in \mathcal{K}(\boldsymbol{S})} \Bigg\{ & \frac{1}{T} \sum_{t=1}^{T} \left\| \boldsymbol{y}_t - \boldsymbol{\beta}\, \boldsymbol{x}_t \right\|_2^2 \;+\; \lambda\,(1-\alpha) \sum_{g=1}^{G} q_g \sum_{k \in \mathcal{C}_g} A_k \left\| \boldsymbol{\beta}_{k,\cdot} - \boldsymbol{\beta}^0_{k,\cdot} \right\|_2 \\ & +\; \lambda\,\alpha \sum_{k=1}^{N} \sum_{j=1}^{M} W_{kj} \left| \beta_{kj} - \beta^0_{kj} \right| \Bigg\}, \end{aligned} \tag{6}$$

where $\lambda \geq 0$ is the penalty strength, $\alpha \in [0,1]$ mixes the group and cell penalties, and $\boldsymbol{\beta}^0$ is a specified target matrix, zero unless stated otherwise. The group multiplier is $q_g = \sqrt{|\mathcal{C}_g|/G}$. The software also offers the $q_g = \sqrt{|\mathcal{C}_g|}$ convention, and experiments must identify a change from the normalized convention used here. The HCGL group norm runs over predictors within a response. Conditional on the partition, signs and weights, its objective and constraints separate across responses. Clusters affect shared preprocessing and the multiplier, not a joint response norm. With $\alpha = 1$ the estimator is a sign-constrained Lasso (Tibshirani 1996) with cluster-pooled signs, which is the case the learned-group tuning selects most often (Section 4.2).

The factor-clustering group LASSO (FCGL) instead minimizes the same loss and cell penalty with group term

$$\lambda(1-\alpha) \sum_{g=1}^{G} q_g \sum_{j=1}^{M} A_{gj} \left\| \boldsymbol{\beta}_{\mathcal{C}_g,j} - \boldsymbol{\beta}^0_{\mathcal{C}_g,j} \right\|_2. \tag{7}$$

This norm couples responses within each cluster–predictor block. HCGL and FCGL use the same gate but are different fitted estimators.

With zero target and no sign restrictions, FCGL is the disjoint-block special case of the multivariate sparse group LASSO objective in Li et al. (2015, Equation (2)). Set their coefficient matrix to $\boldsymbol{B} = \boldsymbol{\beta}^\top$ and their groups to $\mathcal{G} = \{\boldsymbol{B}_{j,\mathcal{C}_g} : j = 1, \ldots, M,\ g = 1, \ldots, G\}$. Dividing our objective by two gives their $1/(2T)$ loss normalization, with cell penalties $\lambda_{jk} = \lambda\alpha W_{kj}/2$ and group penalties $\lambda_{(g,j)} = \lambda(1-\alpha)q_g A_{gj}/2$. For singleton blocks the norm is an absolute value and its penalty can be absorbed into the cell penalty. The unconstrained FCGL comparator therefore implements this group structure within their objective family, without their more general overlapping or nested group constructions or their original tuning protocol. Our additional operation is the construction of restrictions from pooled marginal evidence.

All fitted comparisons use $\boldsymbol{\beta}^0 = 0$ and $W_{kj} = A_k = A_{gj} = 1$. The optional target and adaptive-weight construction is specified in Online Resource 1, Section S14, and is not validated by these experiments. We claim no support-recovery or oracle distribution for the fitted estimators. Online Resource 1, Section S12, gives the implementation and numerical checks.

3 On-support gate signs and pooling precision

The result below concerns the sign matrix produced by the gate before coefficient estimation. Let $(\boldsymbol{x}_t, \boldsymbol{y}_t)$ denote population-centered observations with $\boldsymbol{y}_t = \boldsymbol{\beta}_{\text{true}}\boldsymbol{x}_t + \boldsymbol{\varepsilon}_t$. For a nonempty response subset $\mathcal{C}$ with $m = |\mathcal{C}|$ and $\bar{y}_{t,\mathcal{C}} = m^{-1}\sum_{k\in\mathcal{C}} y_{tk}$, define the population pooled slope, its marginal residual and its score scale,

$$b_{j,\mathcal{C}} = \frac{E[x_{tj}\bar{y}_{t,\mathcal{C}}]}{E[x_{tj}^2]}, \qquad \bar{r}_{t,j,\mathcal{C}} = \bar{y}_{t,\mathcal{C}} - b_{j,\mathcal{C}}x_{tj}, \qquad \omega^2_{j,\mathcal{C}} = \frac{E[x_{tj}^2\bar{r}^2_{t,j,\mathcal{C}}]}{E[x_{tj}^2]^2}. \tag{8}$$

A pair $(j,\mathcal{C})$ is relevant if $\beta_{\text{true},kj} \neq 0$ for at least one $k \in \mathcal{C}$. Let $\mathfrak{C}$ be a deterministic finite family of response subsets. The conditions are:

(A1) The dates $(\boldsymbol{x}_t, \boldsymbol{\varepsilon}_t)$ are iid with $E[\boldsymbol{x}_t] = \mathbf{0}$ and $E[\boldsymbol{\varepsilon}_t \mid \boldsymbol{x}_t] = \mathbf{0}$. The matrix $\boldsymbol{\beta}_{\text{true}}$ and the dimensions N, M do not vary with T. Response errors within a date may be dependent.
(A2) All components of $\boldsymbol{x}_t$ and $\boldsymbol{\varepsilon}_t$ have finite fourth moments, $\boldsymbol{\Sigma}_F = E[\boldsymbol{x}_t\boldsymbol{x}_t^\top]$ is positive definite, and standardized columns have positive population variance. Only complete observations are used.
(A3) The supplied or estimated partition $\widehat{\mathcal{P}}_T$ satisfies $\Pr(\widehat{\mathcal{P}}_T \subseteq \mathfrak{C}) \to 1$, and within every $\mathcal{C} \in \mathfrak{C}$ all nonzero $\beta_{\text{true},kj}$ share one sign for each j.
(A4) For constants $\delta > 0$ and $0 < c < C < \infty$, every relevant pair with $\mathcal{C} \in \mathfrak{C}$ has $|b_{j,\mathcal{C}}| \geq \delta$ and $c \leq \omega^2_{j,\mathcal{C}} \leq C$.
(A5) The gate uses the complete-data, equal-weight statistic of Equation (4) with a deterministic threshold $\tau_T \geq 0$, $\tau_T = o(\sqrt{T})$.
(A6) For every relevant $(j,\mathcal{C})$ with $\mathcal{C} \in \mathfrak{C}$ and every nonzero $\beta_{\text{true},kj}$ in that cluster, $\operatorname{sign}(b_{j,\mathcal{C}}) = \operatorname{sign}(\beta_{\text{true},kj})$.

Condition (A3) is an assumption about the clustering procedure under the data-generating law. Corollary 1 in Online Resource 1, Section S1, supplies sufficient

population Ward and cut-margin conditions for it, and a coherent Gaussian block model satisfies them. Condition (A6) is the assumption that carries the result, and the lemma below makes it checkable.

Proposition 1 (correct gate signs on the support) *Let $\widehat{\boldsymbol{S}}_T$ be the sign matrix of Equation (3) with no unrestricted cells. Under (A1)–(A6),*

$$\Pr\Big\{\widehat{S}_{T,kj} = \operatorname{sign}(\beta_{\mathrm{true},kj}) \text{ for every } \beta_{\mathrm{true},kj} \neq 0\Big\} \longrightarrow 1. \tag{9}$$

The proof is in Online Resource 1, Section S1. Its steps are a law of large numbers for the pooled slope and for $T\widehat{V}_{j,\mathcal{C}}$, whose limit $\omega^2_{j,\mathcal{C}}$ is positive by (A4), and a union bound over the finite family $\mathfrak{C}$, which controls every admissible cluster simultaneously and so allows clusters estimated from the same sample. The proposition does not assert that $\operatorname{supp}(\widehat{\boldsymbol{\beta}}_T)$ equals the true support, provides no oracle distribution for the penalized estimator, and is not a high-dimensional result.

Under (A1) and (A2) the population pooled slope is

$$b_{j,\mathcal{C}} = \frac{1}{m\,\Sigma_{F,jj}} \sum_{k\in\mathcal{C}} \sum_{\ell=1}^{M} \beta_{\mathrm{true},k\ell}\Sigma_{F,j\ell}, \tag{10}$$

so a marginal sign aggregates the own effects $\beta_{\mathrm{true},kj}$ with leakage from every other predictor correlated with $\boldsymbol{x}_j$. The lemma bounds the leakage.

Lemma 1 (a sufficient condition for marginal-sign agreement) *Assume (A1), (A2) and the coherence part of (A3) for a relevant pair $(j,\mathcal{C})$, with common sign $s \in \{-1,+1\}$ on its nonzero $\beta_{\mathrm{true},kj}$. If*

$$\Sigma_{F,jj} \sum_{k\in\mathcal{C}} |\beta_{\mathrm{true},kj}| \;>\; \sum_{k\in\mathcal{C}} \sum_{\ell\neq j} |\beta_{\mathrm{true},k\ell}|\,|\Sigma_{F,j\ell}|, \tag{11}$$

then $\operatorname{sign}(b_{j,\mathcal{C}}) = s$ and $|b_{j,\mathcal{C}}|$ is at least the difference of the two sides divided by $m\,\Sigma_{F,jj}$. Condition (A6) holds for every relevant pair that satisfies Equation (11), and the slope part of (A4) holds with δ equal to the smallest of these lower bounds.

The proof is a triangle inequality applied to Equation (10) and is given in Online Resource 1, Section S1. With uncorrelated predictors the right-hand side is zero and the condition holds for every relevant coherent pair. With correlated predictors it requires the cluster-summed own effect to dominate the cluster-summed leakage, a condition of the same type as, but distinct from, the irrepresentable condition for Lasso sign consistency of Zhao and Yu (2006) and the screening conditions of Fan and Lv (2008). The condition can be evaluated on any simulation design. It cannot be verified on the eQTL data, where $\boldsymbol{\beta}_{\mathrm{true}}$ is unknown.

Pooling precision can be calculated separately for a prespecified cluster. For this variance calculation the analysis scales are fixed in advance. Write $a_{kj} = \operatorname{Cov}(x_{tj}, y_{tk})/\operatorname{Var}(x_{tj})$ for its member marginal slopes and $r^{(j)}_{tk} = y_{tk} - a_{kj}x_{tj}$ for the

population-centered marginal residuals. Their average slope is $b_{j,\mathcal{C}} = m^{-1}\sum_{k\in\mathcal{C}} a_{kj}$, where $m = |\mathcal{C}|$. Under linear conditional means, independent dates and conditional residual covariance $\mathrm{Cov}(\boldsymbol{r}^{(j)}_{t,\mathcal{C}} \mid x_{tj}) = \boldsymbol{\Omega}_j$ constant over dates, centered ordinary least squares (OLS) gives

$$\mathrm{Var}(\widehat{b}_{j,\mathcal{C}} \mid \boldsymbol{x}_j) = \frac{\mathbf{1}^\top \boldsymbol{\Omega}_j \mathbf{1}}{m^2\, \boldsymbol{x}_j^\top \boldsymbol{x}_j}. \tag{12}$$

This follows by writing the slope error as $(m\,\boldsymbol{x}_j^\top \boldsymbol{x}_j)^{-1}\sum_t x_{tj}\sum_{k\in\mathcal{C}} r^{(j)}_{tk}$ and summing independent date variances. Heterogeneous member slopes do not add a separate variance term to their average, but they do change the signal being estimated.

If the marginal residual variances are all σ_j^2 and their correlations are all ρ_e, Equation (12) becomes

$$\frac{\sigma_j^2\{1 + (m-1)\rho_e\}}{m\,\boldsymbol{x}_j^\top \boldsymbol{x}_j}. \tag{13}$$

When member slopes also have the same magnitude and direction, the ratio of pooled to per-response signal-to-noise is $\sqrt{m/\{1 + (m-1)\rho_e\}}$. It is $\sqrt{m}$ for independent marginal residuals and one for identical residuals. For fixed m, both procedures have root-T order, and the ratio is a precision constant. Unequal effects, unequal variances or partial support sharing need not produce this gain and can dilute the pooled signal.

Fig. 1 checks Equation (13) and the two variance estimators by Monte Carlo. One predictor with $T = 80$ is held fixed, $m \in \{1, 2, 4, 8, 16\}$ responses share the slope 0.2, and their residuals are equicorrelated with $\rho_e \in \{0, 0.3, 0.6, 0.9\}$, over 20,000 replications per cell (Online Resource 1, Section S2). Panel (a) shows that the empirical ratio of member to pooled standard errors follows the formula at every cell, with a largest deviation of 0.010 at $m = 2$, $\rho_e = 0$ (1.424 against 1.414). Panel (b) shows that the date-score variance of Equation (4) averages between 0.959 and 0.995 of the true conditional variance, whereas the independent-cell comparator averages $1/\{1 + (m-1)\rho_e\}$ of it, falling to 0.068 at $m = 16$, $\rho_e = 0.9$. At threshold 2 the pooled gate retains the active slope in 0.95 of replications at $m = 4$, $\rho_e = 0$, against 0.44 for the singleton gate, and in 0.58 at $\rho_e = 0.6$. The precision gain is real, and it shrinks quickly with the marginal-residual correlation.

The covariance in this calculation is that of marginal regression residuals, not of the full-model errors or the raw responses. Even independent full-model errors can leave correlated marginal residuals through omitted predictors. Equation (4) aggregates empirical marginal-residual scores without replacing them by a raw-response correlation adjustment. Its positive score-scale limit supports Proposition 1, including the specified sample standardization. A fixed gate threshold does not provide multiplicity control, and dependence-aware standard errors cannot correct a marginal sign that differs from the corresponding partial sign.

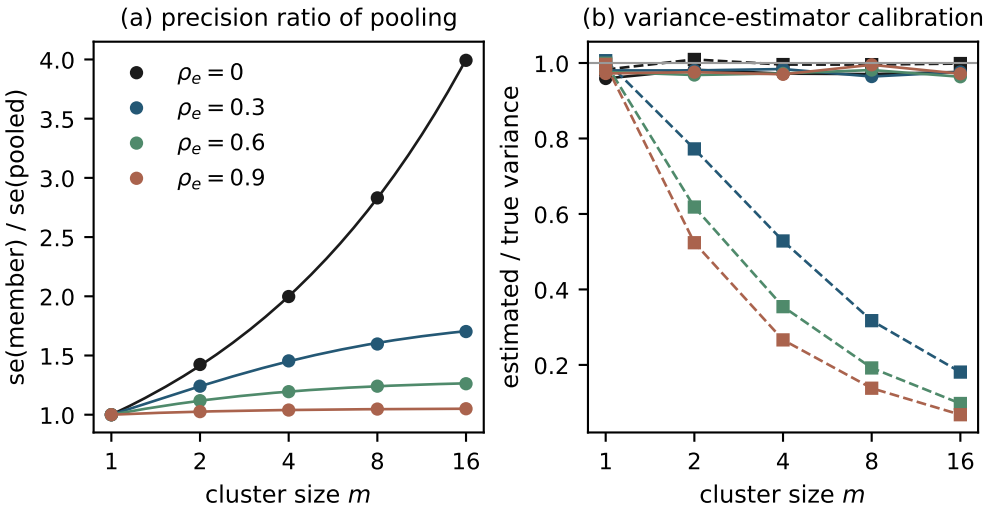


Fig. 1 Pooling precision follows $\sqrt{m/\{1+(m-1)\rho_e\}}$, and the independent-cell variance overstates precision by $1+(m-1)\rho_e$. Monte Carlo with one fixed predictor, $T = 80$, common member slope 0.2, equicorrelated unit-variance marginal residuals and 20,000 replications per cell. (a) Ratio of the member-slope standard error to the pooled-slope standard error, Monte Carlo (points) against Equation (13) (lines). (b) Mean of the date-score variance (circles) and of the independent-cell variance (squares), each divided by the Monte Carlo variance of the pooled slope. Produced by `pooling_precision_check.py`.

4 Simulation study

We distinguish preliminary gate signs, fitted selection and prediction. Both studies use independent test samples and training-only validation. Online Resource 1 gives complete protocols and all coefficient, support and sensitivity results. Intervals describe paired Monte Carlo variation. They do not adjust for multiple comparisons or the disclosed grid extensions.

4.1 Fitted-estimator benchmark

Six designs each use 20 paired seeds, 80 training and 1,000 test observations, 12 predictors and 12 responses. The baseline has three equal clusters, shared supports and signs, independent Gaussian predictors/errors and population $R^2 = 0.30$. Variants introduce weaker signals, partial supports, mixed signs, correlated predictors/errors, or unequal groups and error scales. Four matched HCGL variants differ in their restrictions (unconstrained, singleton/exclude, pooled/exclude and pooled/release). Comparators are unconstrained FCGL, cooperative Lasso, ordinary Lasso and full UniLasso with its leave-one-out univariate features and nonnegative second stage, and a known-group HCGL fit is an oracle diagnostic. HCGL/FCGL fix $\alpha = 0.5$ and the estimated partition uses cut 0.5. Lasso selects one common penalty fraction across responses rather than tuning each response separately. Three-fold training validation repeats preprocessing and jointly selects the penalty and applicable threshold from fourteen penalty fractions of each method's training-only zero-solution scale and the signed thresholds

$\{0, 0.75, 1.5, 2.5, 3, 6, \infty\}$. The grids were extended after inspecting boundary selections, using the same seeds and folds, and no test outcome or coefficient truth selects a candidate. Online Resource 1, Sections S3 and S4, specify every design and report every result, including independent-test mean squared errors (MSE) in Table S2.

This benchmark barely pools. The estimated partitions have a mean of 10.15 groups in the baseline, 12 under weak signal and 3 in the correlated case, out of 12 responses. Pooling has no general predictive advantage over unconstrained HCGL. In the correlated case its paired MSE loss is 0.01568 (Monte Carlo standard error 0.00641, paired t_{19} interval $[0.00226, 0.02909]$), and releasing failed gates gives MSE 0.795 against 0.797 unconstrained and 0.812 with exclusion. Known groups help in the baseline but harm prediction under mixed signs. The correlated case also shows why fitted flip rates alone mislead. Excluded HCGL has zero observed fitted sign flips, yet pins 25.6% of active coefficients to zero and assigns an incorrect gate direction to 12.8% of active cells. Synthetic support summaries use tolerance 10^{-4} in standardized units, following an independent solver audit. Changing from 10^{-6} alters some recovery proportions by up to five percentage points, and leaves prediction and unthresholded coefficient error unaffected.

4.2 Learned pooling in identifiable response groups

To examine genuinely learned groups, five designs each use 40 paired seeds, 80 training observations, 1,000 independent test observations, 12 predictors and three response blocks of size four. Two strong shared anchor predictors and a weaker third predictor create a favorable population signal share of 0.65. Mixed-direction and partial-support designs alter the third effect, and two further designs add predictor or within-block error correlation. This is a diagnostic grouped regime, not an estimate of the yeast signal strength. Online Resource 1, Section S5, gives the construction.

The eight feasible methods use zero targets and nonadaptive weights. Three-fold inner validation jointly selects all applicable combinations of cut $\{0, 0.175, 0.35, 0.5, 0.7, 0.85, 1\}$, mixing $\{0, 0.1, 0.25, 0.5, 0.75, 1\}$, threshold $\{0, 0.75, 1.5, 3, 6, \infty\}$ and the fourteen penalty fractions. Lasso/UniLasso tune only their penalty, and cooperative Lasso also tunes the cut. These final grids include natural domain endpoints and followed two disclosed extensions after boundary diagnostics. All preprocessing is fitted inside training, and coefficient truth is used only for evaluation. A fresh cross-validation reproduction recomputes every candidate loss. Finite grid resolution and selection variability remain.

A paired reference fixes $(\kappa, \alpha, \tau) = (0.5, 0.5, 1.5)$ and tunes only the penalty. One-factor sensitivity changes one setting at a time. At this cut the coherent groups are recovered exactly (mean adjusted Rand index one), and mixed directions give mean index 0.617 and 5.075 learned groups. Group recovery and suitability for a common sign restriction are different targets, and jointly tuned procedures may select different partitions.

Table 1 reports prediction for all eight procedures.

Pooled exclusion improves prediction relative to the corresponding jointly tuned unconstrained HCGL procedure in the coherent, mixed, partial-support and residual-correlation designs. The coherent paired MSE difference is -0.0129 (95% interval

Table 1 Pooled exclusion raises coherent-design test R^2 from 0.614 to 0.626, level with cooperative Lasso and UniLasso. Independent-test R^2 for jointly tuned procedures, means (Monte Carlo standard errors) over 40 paired seeds per design.

Method	Coherent	Mixed	Partial	Predictor corr.	Residual corr.
Lasso	0.614 (0.002)	0.614 (0.002)	0.614 (0.002)	0.612 (0.002)	0.616 (0.002)
HCGL, unconstrained	0.614 (0.002)	0.613 (0.002)	0.614 (0.002)	0.612 (0.002)	0.616 (0.002)
HCGL, singleton/exclude	0.617 (0.002)	0.616 (0.002)	0.617 (0.002)	0.607 (0.005)	0.618 (0.003)
HCGL, pooled/exclude	0.626 (0.002)	0.619 (0.002)	0.619 (0.002)	0.613 (0.005)	0.621 (0.003)
HCGL, pooled/release	0.617 (0.002)	0.613 (0.002)	0.616 (0.002)	0.617 (0.002)	0.617 (0.002)
FCGL, unconstrained	0.623 (0.002)	0.623 (0.002)	0.618 (0.002)	0.623 (0.002)	0.621 (0.002)
Cooperative Lasso	0.624 (0.002)	0.620 (0.002)	0.619 (0.002)	0.622 (0.002)	0.621 (0.002)
UniLasso	0.623 (0.002)	0.622 (0.002)	0.624 (0.002)	0.603 (0.005)	0.621 (0.003)

$[-0.0156, -0.0101]$), and the predictor-correlation contrast includes zero (Online Resource 1, Table S9). Cooperative Lasso, unconstrained FCGL and UniLasso remain level with pooled exclusion. FCGL has the highest mean prediction score in the mixed-direction and predictor-correlation designs, and UniLasso has the highest mean in the partial-support design. The final procedure selects $\alpha = 1$ in 32–40 of 40 pooled-exclusion replications per design. With a positive penalty these are pure elementwise L1 fits, so the fitted estimator in this study is a sign-constrained Lasso with cluster-pooled gates, and the comparison does not establish an advantage for the group-penalty term.

The fixed-reference comparisons expose the cost of pooling incompatible cells. Relative to singleton screening at identical cut, mixing and threshold settings, pooled exclusion changes standardized test MSE by 0.01547 (95% confidence interval $[0.00835, 0.02260]$) under mixed directions and 0.02925 ($[0.02291, 0.03559]$) under partial support. Joint tuning partly avoids these settings, often by selecting a finer partition. Recovery of the designed anchor groups and suitability of a partition for sign screening are distinct criteria.

Gate tradeoffs at a common null-retention level. We separately evaluate singleton and learned-pooled gates at fixed cut 0.5 over thresholds from zero to eight in increments of 0.5. Fig. 2 draws every seed's curve of correct active signs against null-cell retention together with the mean curve. For each seed, linear interpolation compares correct active signs at a common null-cell retention of 0.10. This interpolation uses coefficient truth and is an oracle diagnostic, not a practical threshold calibration rule or a fitted estimator comparison. Both finite curves bracket the target in all 40 seeds of every design, and coverage is also 40/40 at the additional targets 0.05, 0.20 and 0.40 saved with the results.

Table 2 gives the common-null-retention comparison.

At this operating point, pooling improves correct-sign recovery in the coherent, predictor-correlation and residual-correlation designs, but loses recovery under mixed directions and partial support. Recognizable response groups alone therefore do not give pooled screening a uniformly preferable error–coverage tradeoff. The predictor-correlation gate gain also does not establish a prediction gain for hard-exclusion fitting.

Cut and threshold sensitivity is substantial. At the fixed reference settings, raising the cut from 0.5 to 0.7 increases mixed-direction test MSE by 0.0683 (95% interval $[0.0577, 0.0789]$), whereas decreasing it to 0.35 reduces MSE by 0.0187

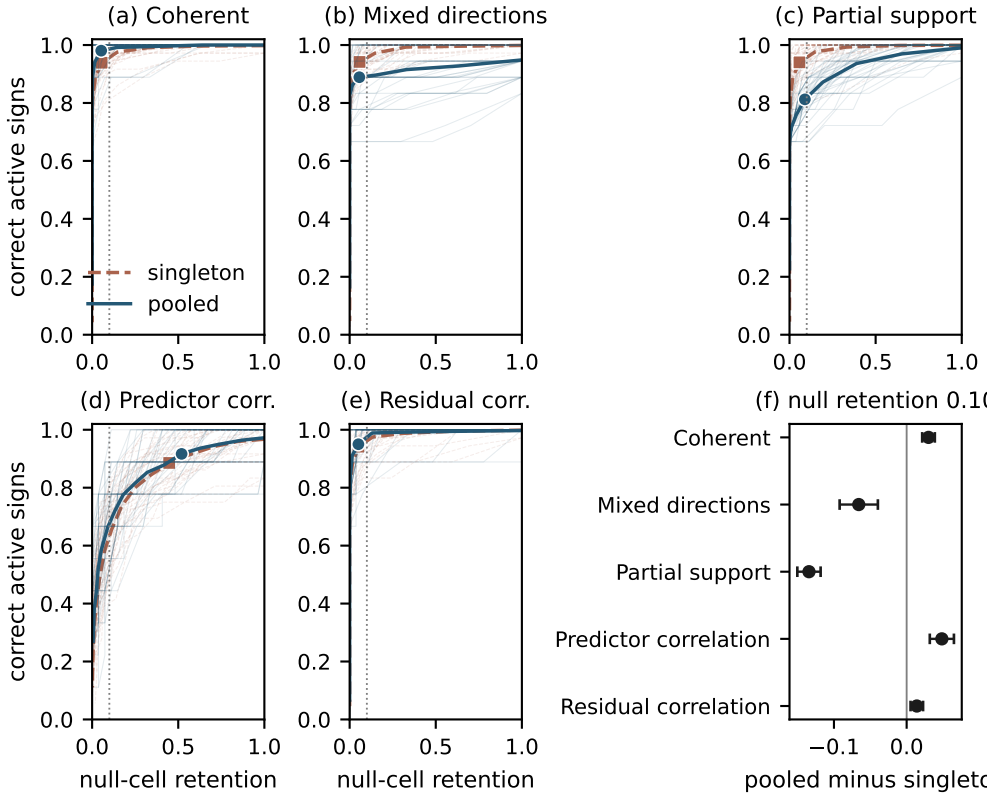


Fig. 2 Pooling helps when cluster members share supports and directions and harms otherwise. (a)–(e) Proportion of correct active signs against null-cell retention for the singleton (dashed) and learned-pooled (solid) gates at cut 0.5, thresholds from zero to eight in steps of 0.5, five designs of the learned-group study with 80 training observations. Thin lines are the 40 seeds and thick lines their means at each threshold, with the threshold-2 operating point marked. The dotted line is null-cell retention 0.10. (f) Paired pooled minus singleton difference in correct active signs at null retention 0.10 with 95% paired t_{39} intervals, the entries of Table 2. Produced by `main_text_figures.py` from the saved gate curves.

Table 2 At null-cell retention 0.10 pooling gains 0.030 correct active signs under coherent groups and loses 0.066 and 0.134 under mixed directions and partial support. Oracle equal-null diagnostic. Correct active-sign proportions are means (Monte Carlo standard errors), and differences are pooled minus singleton with paired t_{39} intervals. The gate cut is fixed, independently of the fitted-estimator joint-CV choices.

Design	Singleton	Pooled	Difference [95% CI]	Seeds
Coherent	0.958 (0.007)	0.989 (0.005)	0.0301 [0.0214, 0.0388]	40/40
Mixed directions	0.958 (0.006)	0.892 (0.015)	-0.0659 [-0.0922, -0.0396]	40/40
Partial support	0.959 (0.007)	0.824 (0.010)	-0.1344 [-0.1504, -0.1183]	40/40
Predictor correlation	0.647 (0.018)	0.695 (0.021)	0.0484 [0.0318, 0.0651]	40/40
Residual correlation	0.960 (0.007)	0.974 (0.009)	0.0140 [0.0054, 0.0226]	40/40

([−0.0266, −0.0109]). The final joint search selects cut 0.35 in 26 of 40 mixed-direction and 21 of 40 partial-support replications, and selects the singleton endpoint in another 8 and 11, respectively. Online Resource 1, Section S6.1, reports selected one-factor contrasts and tuning frequencies, and Section S6 reports the coefficient and support outcomes.

5 Application to yeast eQTL data

Expression quantitative trait loci (eQTL) relate genetic markers to gene expression. The yeast data provide many correlated responses with uncertain grouping and more candidate predictors than observations, and the same cross is the benchmark of Yin and Li (2011), Kim and Xing (2012) and Li et al. (2015). This makes them useful for examining the practical consequences of marginal sign restrictions. The partial-coefficient signs are unknown, so this application cannot verify the structural assumptions of Proposition 1. We evaluate prediction and resampling stability of the complete pipeline without treating marginal signs as conditional coefficient truth.

5.1 Data and evaluation scope

The cross of two yeast strains provides $T = 112$ segregants (Brem and Kruglyak 2005). Membership in the Kyoto Encyclopedia of Genes and Genomes (KEGG) mitogen-activated protein kinase (MAPK) pathway gives $N = 64$ genes, rather than the 54-gene subset used in Yin and Li (2011). The annotations include pheromone response, filamentation, high-osmolarity response, cell-wall integrity and a sporulation (Smk1) category. The original panel contains 3244 markers. The primary predictive comparison uses the nested protocol below.

5.2 Repeated nested validation

The protocol starts from all 3,244 markers and the 64 genes selected by the prespecified MAPK annotation, and uses three repeats of three outer folds with three inner folds (seed 20261024). Each inner training fold independently recomputes linkage-disequilibrium (LD) thinning at absolute correlation 0.92, marginal screening at $p \leq 0.01$ in at least two genes, centering, scaling, Ward clustering and gate signs. The selected procedure is refitted using only the outer training observations. Outer outcomes are used only for prediction assessment and the separate descriptive diagnostics below. Outer training sets retain 136–194 markers and form three or four response clusters.

All observations receive equal weight. The date-score threshold $\tau = 2$, final-merge cut fraction 0.7, mixing parameter $\alpha = 0.5$ for HCGL/FCGL, zero prior target and nonadaptive weights $W = 1$ are fixed in advance. These are fixed reference settings for matched comparisons, and neighboring settings are assessed in Section 5.5. Within each inner fold, each method has its own zero-solution Karush–Kuhn–Tucker (KKT) penalty scale $\lambda_{\max}$, respecting its groups and restrictions. The candidate fractions are

$\{1, 0.3, 0.1, 0.03\}\lambda_{\max}$. We minimize pooled inner validation squared error in training-standardized response units. Every outer fit selects fraction 0.3, an interior grid point, and the corresponding numerical penalties differ across methods.

The eight methods comprise ordinary unconstrained Lasso, singleton-gated Lasso, unconstrained HCGL, singleton-gated HCGL, pooled HCGL with exclusion or release, and unconstrained or pooled-excluded FCGL. Release retains the half-space for a passed gate but leaves a failed gate unrestricted. The four HCGL variants share the row-norm penalty, so the singleton-versus-pooled HCGL comparison isolates the gate's pooling. The unrestricted FCGL comparison isolates restrictions in the coupled penalty. Both additional baselines use the same outer/inner splits, training-only preprocessing and penalty-fraction grid.

For gene k and repeat r, predictions from all outer test folds are pooled before computing

$$R^2_{kr} = 1 - \frac{\sum_i (y_{ik} - \widehat{y}_{ik}^{(-f(i),r)})^2}{\sum_i (y_{ik} - \overline{y}_k^{(-f(i),r)})^2},$$

where the baseline mean is learned from the corresponding outer training fold. We report the median and mean over genes, and pooled R^2, which sums both numerator and denominator over genes before taking their ratio. The pooled score gives more weight to genes with larger baseline squared variation in their original expression units.

Table 3 reports the three predictive summaries and Fig. 3 the per-gene distributions behind them.

Table 3 At fixed settings pooled exclusion lowers median-gene R^2 from 0.073 to 0.035 and release leaves it at 0.073. Nested eQTL prediction with $\kappa = 0.7$, $\tau = 2$ and four penalty fractions. Entries are means across three repeats, and parentheses are the standard deviations across those splits of the same 112 segregants. They are descriptive split variability, not sampling standard errors or confidence intervals.

Method	Median gene R^2	Mean gene R^2	Pooled R^2
Unconstrained Lasso	0.051 (0.004)	0.136 (0.005)	0.534 (0.007)
Singleton-gated Lasso, exclude	0.051 (0.004)	0.136 (0.005)	0.534 (0.007)
HCGL, unconstrained	0.073 (0.006)	0.136 (0.010)	0.488 (0.016)
HCGL, singleton/exclude	0.065 (0.012)	0.131 (0.009)	0.486 (0.015)
HCGL, pooled/exclude	0.035 (0.010)	0.100 (0.012)	0.398 (0.047)
HCGL, pooled/release	0.073 (0.005)	0.134 (0.011)	0.473 (0.038)
FCGL, unconstrained	0.056 (0.009)	0.130 (0.004)	0.498 (0.006)
FCGL, pooled/exclude	0.036 (0.002)	0.114 (0.007)	0.419 (0.043)

Hard pooled exclusion lowers median-gene R^2 from 0.073 for unconstrained HCGL to 0.035, and releasing failed gates gives 0.073. These data therefore do not support a predictive benefit from hard exclusion. The matched singleton-exclusion HCGL has median-gene $R^2 = 0.065$. Pooled minus singleton exclusion has paired median-gene difference -0.0307 (split SD 0.0045), negative in all three repeats. FCGL exclusion likewise reduces median-gene R^2 from 0.056 to 0.036, a paired difference of -0.0195 (split SD 0.0068), also negative in every repeat. These contrasts separate the effects

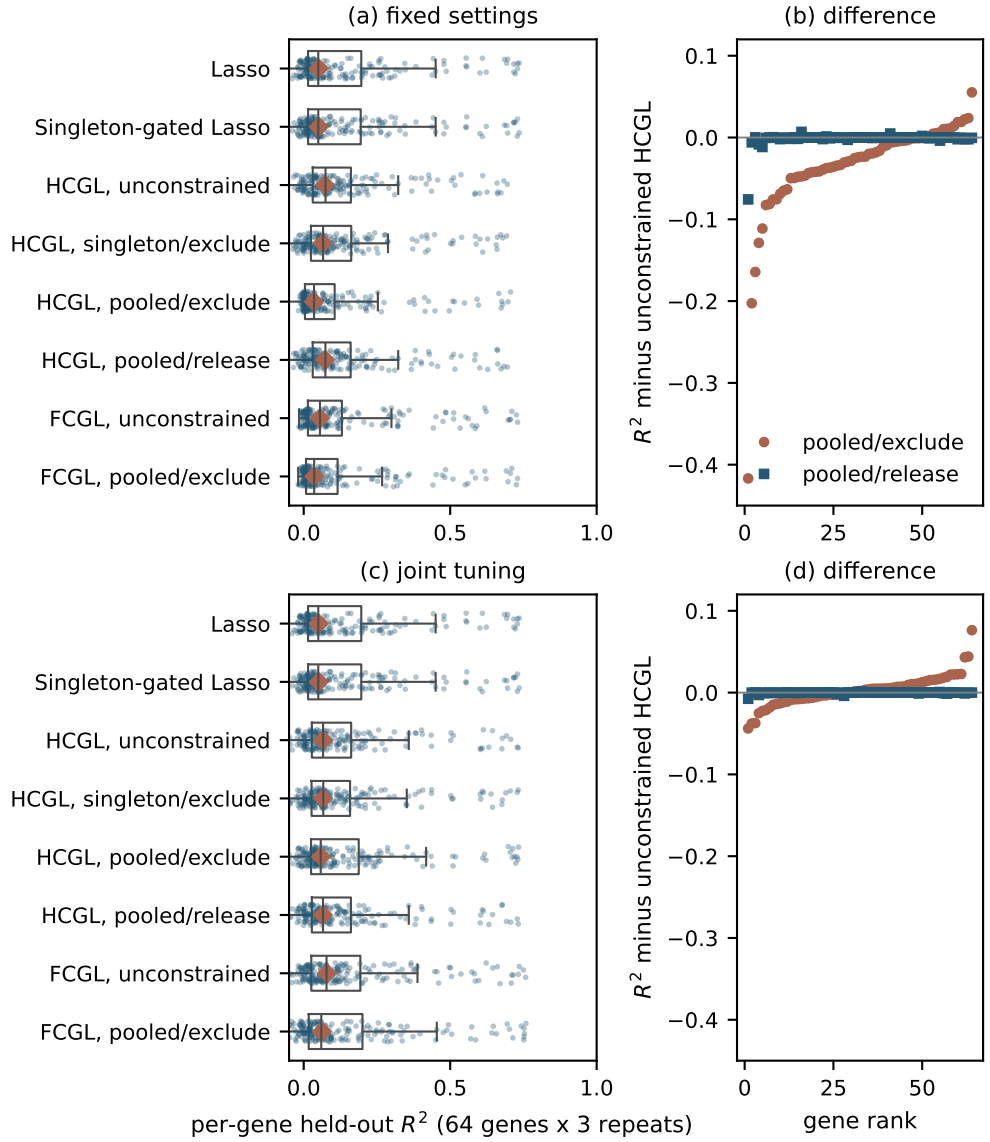


Fig. 3 At fixed settings the loss from pooled exclusion is spread across genes, and joint tuning removes it. (a), (c) Held-out R^2 for each of the 64 genes in each of the three repeats (points and boxes, whiskers at 1.5 interquartile ranges, outliers shown as points) for the eight nested procedures, with the median-gene statistics of Tables 3 and 4 (diamonds). (b), (d) Per-gene difference in R^2, averaged over repeats, of pooled exclusion (circles) and pooled release (squares) against unconstrained HCGL, sorted by the exclusion difference. The upper row is the fixed-setting protocol and the lower row the jointly tuned protocol of Section 5.3. Produced by `eqtl_joint_report.py` from the saved nested predictions.

of learned pooling and hard restrictions from the choice of row or coupled penalty. Fig. 3(b) shows that the exclusion loss at these settings is broad rather than concentrated. Averaged over repeats, exclusion lowers R^2 for 50 of the 64 genes, with a mean per-gene difference of -0.0355, whereas release lowers it for 34 genes with a mean difference of -0.0017. Singleton-gated Lasso matches unconstrained Lasso to three decimals in every summary, so at the singleton level the gate at $\tau = 2$ is inert on these data and only pooling changes the fit. Lasso has lower median-gene R^2 (0.051) than unconstrained HCGL but higher pooled R^2 (0.534 versus 0.488), and its mean-gene score is marginally higher. There is no uniform ranking across these summaries. These results fix the cut and threshold and use a four-point penalty grid, whereas the simulations tune cut and threshold jointly. The next subsection applies the same joint search to these data.

5.3 Joint tuning of cut, threshold and penalty

We repeat the nested validation with the same splits, preprocessing chain and eight estimators, but let each method select its own cut, threshold and penalty by pooled inner validation error. The grid is $\kappa \in \{0.35, 0.5, 0.7, 0.9\}$, $\tau \in \{1, 2, 3\}$ and seven fractions $\{1, 0.5, 0.3, 0.17, 0.1, 0.05, 0.03\}$ of each candidate's training-only $\lambda_{\max}$. Lasso tunes the fraction only, singleton gating the threshold and fraction, the unconstrained HCGL and FCGL fits the cut and fraction, and the pooled and singleton-gated HCGL and FCGL fits all three. Ties choose the earliest grid candidate with the larger fraction. This search was run with the CLARABEL solver. A fixed-setting arm with the archived protocol, run under the same solver on the same splits, reproduces every summary of Table 3 to within 7×10^{-6}, so the solver change does not enter the comparison. Online Resource 1, Section S9, gives the selected settings and the paired contrasts.

Table 4 reports the jointly tuned prediction summaries.

Table 4 With its own cut and threshold, pooled exclusion is level with unconstrained HCGL, and unconstrained FCGL with a tuned cut is the best method. Nested eQTL prediction with cut, threshold and penalty selected per method by inner validation. Entries are means across three repeats and parentheses are split standard deviations, as in Table 3.

Method	Median gene R^2	Mean gene R^2	Pooled R^2
Lasso	0.051 (0.004)	0.137 (0.005)	0.534 (0.007)
Singleton-gated Lasso	0.051 (0.004)	0.137 (0.005)	0.534 (0.007)
HCGL, unconstrained	0.067 (0.009)	0.133 (0.008)	0.503 (0.011)
HCGL, singleton/exclude	0.066 (0.009)	0.132 (0.007)	0.503 (0.011)
HCGL, pooled/exclude	0.059 (0.003)	0.135 (0.007)	0.530 (0.007)
HCGL, pooled/release	0.066 (0.009)	0.132 (0.008)	0.503 (0.011)
FCGL, unconstrained	0.080 (0.005)	0.154 (0.009)	0.543 (0.020)
FCGL, pooled/exclude	0.063 (0.005)	0.142 (0.004)	0.556 (0.005)

Joint tuning changes the pooled gate itself. Pooled exclusion selects $\kappa = 0.35$, $\tau = 1$ and fraction 0.3 in all nine outer fits, which gives 19 to 31 clusters among the 64 genes and the weakest gate on the grid. Its median-gene R^2 rises from 0.035

to 0.059 and its pooled R^2 from 0.398 to 0.530. The paired median-gene difference against unconstrained HCGL falls from -0.0384 (split SD 0.0052, negative in all three repeats) to -0.0073 (split SD 0.0119, negative in two repeats), and the difference against singleton exclusion from -0.0307 to -0.0065. Release selects $\tau = 3$ in four fits and coincides with unconstrained HCGL within 0.001. Unconstrained HCGL itself loses median-gene R^2, from 0.073 to 0.067, because inner validation selects $\kappa = 0.5$ or 0.35 in four of nine fits. The largest change is unconstrained FCGL, whose median-gene R^2 rises from 0.056 to 0.080 with $\kappa = 0.9$ in five fits and 0.5 in three, and whose pooled R^2 of 0.543 exceeds Lasso. FCGL exclusion still trails unconstrained FCGL by 0.0173 (split SD 0.0098, negative in all three repeats) on the median, although it has the highest pooled R^2 of all methods (0.556). Fig. 3(d) shows that the per-gene exclusion differences against unconstrained HCGL now straddle zero, with 28 of 64 genes below and a mean of $+0.0022$.

The joint search therefore removes the exclusion loss by moving the gate toward singletons and a low threshold, which is the same escape the learned-group simulations found, and it does not produce a gain over unconstrained fitting. The grid is coarse, the inner folds are small, and the selected settings vary across splits for every method except pooled exclusion, so the differences between HCGL variants in Table 4 are within split variability.

5.4 Partition and fitted-coefficient stability

We compare all 36 pairs of the nine outer-training fits of the fixed-setting protocol. Cluster adjusted Rand index averages 0.420 (range 0.172–0.647), and selected-marker Jaccard overlap averages 0.253 (range 0.188–0.306). Across these fits, 579 of the 3,244 markers are selected at least once. LD-representative overlap averages 0.612. Both response grouping and predictor preprocessing therefore vary across splits. These are descriptive comparisons of overlapping samples, not 36 independent replications or sampling confidence intervals.

For each gene–marker cell, we distinguish a marker absent after preprocessing, a retained marker with a fitted zero, and positive or negative fitted coefficients. Selection uses magnitude greater than 10^{-6} in training-standardized units. Jaccard overlap divides the number selected in both fits by the number selected in either, and the common-marker version first restricts to markers retained in both training sets. Direction agreement uses only cells selected in both fits. Table 5 shows why these denominators matter. Near-perfect conditional direction agreement accompanies low overall selection overlap. Hard pooled exclusion does not improve selection overlap over the matched unconstrained, singleton-gated or released-gate HCGL fits. Jaccard overlap also depends on selection size and substitution among correlated markers, so a higher value alone does not establish better biological recovery.

Raising the tolerance to 10^{-4} leaves these selection-overlap conclusions essentially unchanged. The largest change across the eight methods is 0.0025, and for pooled exclusion Jaccard remains 0.119 while release gives 0.134. At 10^{-8} numerical residuals affect support and direction classification, so treating every nonzero solver output as selected would be inappropriate. These checks concern resampling stability, not recovery of an unknown biological support.

Table 5 Direction agreement is near one only because selection overlap is low. eQTL fitted-cell stability, means over 36 overlapping split pairs. Jaccard uses all selected cells, Common restricts to jointly retained markers, and Direction conditions on selection in both fits. Shared is the mean number of cells selected in both fits, the denominator of each direction comparison.

Method	Jaccard	Common	Direction	Shared
Unconstrained Lasso	0.111	0.259	1.000	54.9
Singleton-gated Lasso, exclude	0.111	0.265	1.000	75.8
HCGL, unconstrained	0.136	0.382	0.998	584.8
HCGL, singleton/exclude	0.152	0.397	1.000	358.4
HCGL, pooled/exclude	0.119	0.279	1.000	172.9
HCGL, pooled/release	0.134	0.379	0.998	574.7
FCGL, unconstrained	0.099	0.186	0.999	75.6
FCGL, pooled/exclude	0.095	0.183	1.000	52.3

LD thinning compares consecutive genomic markers with the current representative, so it does not bound every remaining pairwise correlation. Among outer-training selected markers, the mean fold median absolute correlation is 0.088, the mean 95th percentile is 0.306, and the mean maximum is 0.969. On average 2.56% of pairs exceed 0.5 and 0.89% exceed 0.8. Residual predictor dependence therefore remains relevant to marginal-versus-conditional sign disagreement.

5.5 Application sensitivity

Ten predeclared one-factor neighbors of the fixed-setting protocol vary LD pruning, outcome screening, Ward cut, gate threshold or mixing while retaining the same inner/outer splits. Only the penalty is tuned, and outer outcomes do not select a setting. Pooled exclusion has lower median-gene R^2 than unconstrained and released-gate HCGL in every repeat at every setting, with mean paired losses of 0.020–0.069 against unconstrained HCGL. The loss changes with the cut and threshold, and release stays close to unconstrained HCGL without establishing equivalence. No outer fit selects a penalty endpoint. Online Resource 1, Sections S7 and S8, reports all settings, results and same-fold configuration stability. Held-out marginal coherence exceeds a size-matched random-partition reference, but does not validate conditional directions or biological pathways (Online Resource 1, Section S11.1).

Separately, we compare independent-cell and date-score variance on the same nine outer splits, holding preprocessing, groups and numerical penalties fixed. Switching to date scores newly excludes 32.71% of all coefficient cells, and the median relative loading-matrix change is 53.62% for HCGL and 18.10% for FCGL. Standardized held-out mean squared error increases by 1.39% and 0.56%, respectively. These are fixed-setting sensitivity results without variance-specific retuning, not replacements for the nested comparisons. Online Resource 1, Section S10, gives the protocol, full scores and synthetic application controls.

6 Computation and reproducibility

Conditional on the partition, signs and weights, the HCGL objective of Equation (6) separates across responses, so the N constrained fits can run in parallel, whereas the FCGL block norm couples each cluster's responses. The solver is CVXPY with MOSEK for the fitted-estimator comparisons and the yeast fits, and CLARABEL for the synthetic software-example fits and for independent reference solves of selected objectives. A valid MOSEK license is needed for full reproduction. Measured path timings across four dimensions are reported in Online Resource 1, Section S12.1, separately from nested-validation cost. A fresh Windows CPython 3.12.7 environment with an installed `factorlasso` 1.0.5 wheel reproduced all 2,680 primary synthetic fits, 16,800 sensitivity refits and the nested eQTL coefficients and performance metrics. This is clean-environment reproduction on the same host, not independent-laboratory or cross-platform replication. The jointly tuned eQTL search of Section 5.3 ran in a Linux container with the `factorlasso` 1.0.5 wheel from PyPI and CLARABEL, with per-fold checkpoints and about 16 processor hours of solver time in its inner loops. CLARABEL reported a possibly inaccurate solution for 70 of the roughly 12,000 solves of both arms. The flagged solves were not mapped to candidates, and every selected candidate uses fraction 0.17 or 0.3 of its penalty scale. Online Resource 2 supplies the exact source, frozen inputs, fitted outputs, environment pins and verification receipts, and identifies the later variance comparison and the all-failed infinite-threshold endpoint correction as a separately hashed overlay. The three main-text figures are produced by `pooling_precision_check.py`, `main_text_figures.py` and `eqtl_joint_report.py` from the saved results.

6.1 Use of AI-assisted tools

During preparation the authors used Anthropic Claude and OpenAI Codex for analysis-code design, code drafting, review and debugging, numerical checks, and prose drafting and editing. The authors review the code, analysis and text and are responsible for the scientific content. Tool-generated output does not supply independent validation.

7 Discussion

The paper shows when pooling marginal sign evidence across response clusters helps a subsequent sparse regression and when it harms. Pooling m responses multiplies the precision of a marginal slope by $\sqrt{m/\{1+(m-1)\rho_e\}}$, and the gate then retains the correct direction on every nonzero coefficient under Proposition 1, whose marginal-sign condition Lemma 1 reduces to a bound on coefficient-weighted predictor covariances. The gain is conditional on coherence. At matched null retention the learned-pooled gate recovers 0.030 more correct active signs than the singleton gate when cluster members share supports and directions, and 0.066 and 0.134 fewer under mixed directions and partial supports. The fitted comparisons repeat this pattern. Pooled exclusion raises coherent-design test R^2 from 0.614 to 0.626 and is level with cooperative Lasso,

unconstrained FCGL and UniLasso. On the yeast data exclusion at fixed settings lowers median-gene R^2 from 0.073 to 0.035 while release leaves it at 0.073. With the cut and threshold tuned jointly with the penalty, exclusion selects the finest partition and the lowest threshold on the grid in every fit, the loss falls to 0.007, and unconstrained FCGL with a tuned cut reaches 0.080. Releasing a failed gate is therefore the safe default, and exclusion pays only when the grouping is coherent.

Gate failure can either exclude a coefficient or leave it unrestricted. Those actions answer different questions. Exclusion screens the support, while unrestricted failure declines to constrain the direction. Their consequences must be evaluated through fitted-coefficient error, selected support, sign coverage and prediction, rather than through low flip rates alone. The first simulation family finds small, uncertain gains in its baseline and a clear loss from exclusion in the correlated case. The recognizable-group study recovers nontrivial groups and finds a coherent-group prediction gain over the corresponding jointly tuned unconstrained HCGL procedure. Joint tuning can change the grouping and does not isolate a pure gate effect. Pure-L1 mixing is selected in 32–40 of 40 pooled-exclusion replications, so those comparisons do not establish an advantage for the group-penalty term. At fixed settings the nested eQTL analysis favours singleton gating or releasing failed gates over pooled exclusion within the same HCGL penalty, and its high direction agreement is conditional on a small overlap of selected cells. Joint tuning closes the exclusion gap by moving the pooled gate toward singletons and a low threshold, which is the escape the simulations also take, and it does not turn the gate into a gain.

For prediction, the cut and gate threshold belong inside the same training-only selection procedure as the penalty. The comparison set should include the unconstrained objective, singleton gating and failed-gate release. A fixed threshold such as 2 is a screening setting, not a validated significance level. Sensitivity to cut, threshold, variance convention and numerical zero classification should accompany interpretation of selected coefficients. The paired variance comparison shows that retaining within-date dependence can materially change loadings without improving prediction, and Fig. 1(b) shows why the independent-cell convention cannot be trusted when marginal residuals are correlated. The package defaults are unchanged, and the date-score convention in this study is an explicit choice.

The limitations are these. The on-support result is fixed-dimensional and assumes a supplied threshold sequence. It is not a selection theorem for the fitted estimator, a calibrated multiple-testing procedure, or a consistency result for the joint cross-validation rule. Marginal signs can disagree with partial signs, and no gate statistic corrects that. The simulation families are small, the grids are finite, and the gains are confined to coherent structures. Repeated splits of one yeast cross measure sensitivity to splitting, not replication across independent biological samples, and marginal-sign agreement there is descriptive. The joint eQTL search uses a $4 \times 3 \times 7$ grid, one solver and inner folds of about 25 segregants. The finite-span, missing-history and serially dependent extensions require additional theory. Reliable external groups or independently validated marginal directions remain the most direct settings in which to investigate a practical benefit from pooling.

Statements and Declarations

Competing interests

Both authors are employed by LGT Bank. The bank had no role in the design of the method, the analysis, or the decision to submit the work for publication. The authors declare no other competing interest.

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Author contributions

Artur Sepp: conceptualization, methodology, software, writing the original draft. Mika A. Kastenholz: methodology, review and editing.

Data and code availability

The yeast data originate from Brem and Kruglyak (2005). Online Resource 2 contains the revised replication source, frozen research inputs, fitted results, dependency pins and verification receipts. The maintained software is available at https://github.com/ArturSepp/factorlasso. The historical archive https://doi.org/10.5281/zenodo.21000294 contains earlier version 0.7.2 experiments and is not the source of the revised results. No financial return histories are required. The replication archive excludes solver licenses, and full research fitting requires a valid MOSEK license.

Ethics approval and consent

This study uses simulations and previously published yeast data. No new human-participant or animal experiment was conducted. Consent to participate and consent to publish personal information are not applicable.

Supplementary information

Online Resource 1 (ESM_1.pdf): Mathematical conditions and proofs, the Monte Carlo check of the pooling precision formula, complete simulation protocols, extended fitted and tuning results, eQTL sensitivity and stability, the jointly tuned eQTL protocol with its selected settings and paired contrasts, paired variance-convention comparisons and numerical checks.

Online Resource 2 (ESM_2.zip): Replication source and research inputs, saved fitted results, environment specifications, reproduction commands and verification receipts for the revised analyses, with the later variance-sensitivity evidence, source changes and the record of historical conventions and corrections identified separately.

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